

Riemannian Geometry of Two-Agent Interaction Dynamics

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Abstract

This work develops a time-dependent Riemannian metric for a two-agent interaction system. The interaction space is modeled as a three-dimensional manifold with one temporal and two interaction coordinates, equipped with a block-diagonal metric whose spatial components evolve in time. Using the Levi-Civita connection derived from the metric, we compute the full Ricci tensor and show that curvature is governed by the temporal deformation rate of the interaction metric. A natural tensor

$$K_j^i = \frac{1}{2} G^{ik} \dot{G}_{kj}$$

emerges as the intrinsic interaction flow tensor. We demonstrate that the Ricci curvature components are determined by the acceleration and nonlinear self-coupling of this interaction flow. This establishes a geometrically consistent formulation of interaction dynamics as curvature evolution on a time-dependent manifold, suggesting connections to matrix Ricci flow and dynamical systems theory.

1 Introduction

Mathematical descriptions of interacting systems often rely on dynamical equations defined in Euclidean state spaces. However, when interaction variables mutually constrain one another, the underlying state space may naturally acquire a nontrivial geometric structure. In such cases, curvature provides a coordinate-invariant measure of structural deformation induced by temporal evolution. In this work, we formulate a geometric model for a two-agent interaction system by equipping the interaction space with a time-dependent Riemannian metric. The model consists of one temporal coordinate and two interaction coordinates associated with an active interaction variable. A passive interaction variable is assumed to be linearly constrained by a time-dependent matrix transformation. Rather than prescribing curvature phenomenologically, we derive it directly from the Levi-Civita connection of the constructed metric. This approach ensures full tensorial consistency and allows the curvature to emerge solely from the temporal deformation of the interaction structure. The resulting

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formulation places two-agent interaction dynamics within the framework of time-dependent Riemannian geometry and establishes a bridge between dynamical systems and curvature evolution on low-dimensional manifolds.

2 Construction of the Interaction Metric

We consider two interaction vectors:

$$I_S(t) \in \mathbb{R}^2, \quad I_P(t) \in \mathbb{R}^2,$$

representing active and passive interaction components, respectively. We assume a linear constraint of the form

$$I_P(t) = \gamma(t)I_S(t),$$

where $\gamma(t)$ is a differentiable 2×2 matrix-valued function of time.

2.1 Extended Euclidean Representation

Consider the extended coordinate space

$$(t, I_P^1, I_P^2, I_S^1, I_S^2) \in \mathbb{R} \times \mathbb{R}^2 \times \mathbb{R}^2,$$

equipped with the standard Euclidean line element

$$ds^2 = dt^2 + dI_P^T dI_P + dI_S^T dI_S.$$

This space serves only as a construction device; the physical interaction space will be obtained by imposing the constraint $I_P = \gamma(t)I_S$.

2.2 Metric Induced by the Constraint

Imposing the constraint yields

$$dI_P = \gamma(t)dI_S + \dot{\gamma}(t)I_S dt.$$

However, to construct a metric purely on the interaction coordinates, we consider variations at fixed time when defining the spatial block of the metric. Hence we focus on the induced quadratic form in dI_S at fixed t :

$$dI_P = \gamma(t)dI_S.$$

Substituting into the Euclidean line element gives

$$ds^2 = dt^2 + dI_S^T \gamma(t)^T \gamma(t) dI_S + dI_S^T dI_S.$$

Rearranging terms,

$$ds^2 = dt^2 + dI_S^T (\mathbf{I} + \gamma(t)^T \gamma(t)) dI_S.$$

2.3 Definition of the Interaction Metric

We therefore define the spatial metric tensor

$$G(t) = \mathbf{I} + \gamma(t)^T \gamma(t).$$

By construction:

- $G(t)$ is symmetric,
- $G(t)$ is positive definite for all t ,
- $G(t)$ depends only on time.

3 Curvature Structure of the Interaction Metric

We consider the 3-dimensional manifold $\mathcal{M} = \mathbb{R}_t \times \mathbb{R}_{I_S}^2$ with coordinates $x^\mu = (t, I_S^1, I_S^2)$. The metric is defined as

$$g_{\mu\nu} = \begin{bmatrix} a^2 & 0 \\ 0 & G_{ij}(t) \end{bmatrix},$$

where

- $a > 0$ is a constant scaling factor for the temporal coordinate,
- $G_{ij}(t)$ is a symmetric positive-definite 2×2 matrix,
- G_{ij} depends only on t .

We denote by G^{ij} the inverse matrix of G_{ij} .

Definition 3.1 (Interaction Flow Tensor). Define the tensor

$$K_j^i = \frac{1}{2} G^{ik} \dot{G}_{kj}$$

where $\dot{G}_{kj} = \frac{d}{dt} G_{kj}$.

This tensor measures the rate of change of the interaction metric.

Theorem 3.1 (Non-vanishing Christoffel Symbols). For the metric above, the only non-zero Christoffel symbols are

$$\begin{aligned} \Gamma_{ij}^0 &= -\frac{1}{2a^2} \dot{G}_{ij}, \\ \Gamma_{0j}^i &= \Gamma_{j0}^i = K_j^i. \end{aligned}$$

Proof. The Christoffel symbol is

$$\Gamma_{\mu\nu}^\lambda = \frac{1}{2} g^{\lambda\sigma} (\partial_\mu g_{\nu\sigma} + \partial_\nu g_{\mu\sigma} - \partial_\sigma g_{\mu\nu}).$$

Since:

- $g_{00} = a^2$ (constant),
- $g_{0i} = 0$,
- $g_{ij} = G_{ij}(t)$,

and the metric depends only on t , we obtain:

1. For Γ_{ij}^0 :

$$\Gamma_{ij}^0 = \frac{1}{2}g^{00}(-\partial_0 g_{ij}) = -\frac{1}{2a^2}\dot{G}_{ij}.$$

2. For Γ_{0j}^i :

$$\Gamma_{0j}^i = \frac{1}{2}G^{ik}\partial_0 G_{kj} = \frac{1}{2}G^{ik}\dot{G}_{kj} = K_j^i$$

All other components vanish. □

Theorem 3.2 (Expression for R_{00}). The 00-component of the Ricci tensor is

$$R_{00} = -\text{Tr}(\dot{K}) - \text{Tr}(K^2).$$

Proof. The Ricci tensor is

$$R_{\mu\nu} = \partial_\lambda \Gamma_{\mu\nu}^\lambda - \partial_\nu \Gamma_{\mu\lambda}^\lambda + \Gamma_{\mu\nu}^\lambda \Gamma_{\lambda\sigma}^\sigma - \Gamma_{\mu\lambda}^\sigma \Gamma_{\nu\sigma}^\lambda,$$

For R_{00} :

1. Since $\Gamma_{00}^\lambda = 0$, the first and third terms vanish.

2. The second term:

$$-\partial_0 \Gamma_{0\lambda}^\lambda = -\partial_0 \Gamma_{0i}^i = \partial_0 \text{Tr}(K) = -\text{Tr}(\dot{K}).$$

3. The fourth term:

$$-\Gamma_{0\lambda}^\sigma \Gamma_{0\sigma}^\lambda = -K_j^i K_i^j = -\text{Tr}(K^2).$$

Combining terms:

$$R_{00} = -\text{Tr}(\dot{K}) - \text{Tr}(K^2).$$

□

Theorem 3.3 (Expression for R_{ij}). The spatial components of the Ricci tensor are

$$R_{ij} = \dot{K}_{ij} + K_{ik} K_j^k + K_{ij} \text{Tr}(K).$$

Proof. Using the Christoffel symbols:

- Only terms containing Γ_{ij}^0 and Γ_{0j}^i survive.
- Spatial derivatives vanish since G_{ij} depends only on t , $\partial_k G_{ij} = 0$.

Computing each term in the Ricci definition yields:

1. A time derivative contribution: $\dot{K}_{ij}$.
2. A quadratic term: $K_{ik}K_j^k$.
3. A trace-coupling term: $K_{ij}\text{Tr}(K)$.

Summing all surviving contributions gives the stated result. $\square$

Definition 3.2 (Emotional Quantity Tensor). Let an emotional quantity in a tensor form defines

$$E_{ij} = \dot{K}_{ij} + K_{ik}K_j^k,$$

From this definition, we give that

- R_{00} measures the negative trace of the total interaction acceleration.
- R_{ij} contains both acceleration and collective coupling via $\text{Tr}(K)$.

This establishes a geometrically consistent relationship between curvature and the temporal evolution of the interaction metric.

4 Conclusion

In this work, we constructed a time-dependent interaction metric for a two-agent system,

$$g_{\mu\nu} = \begin{bmatrix} a^2 & 0 \\ 0 & G_{ij}(t) \end{bmatrix},$$

where the spatial block $G_{ij}(t)$ represents the evolving structure of interaction between agents. Instead of imposing a connection a priori, the curvature structure was derived strictly from the Levi-Civita connection associated with the metric. This led naturally to the emergence of the tensor

$$K_j^i = \frac{1}{2}G^{ik}\dot{G}_{kj},$$

which characterizes the temporal deformation rate of the interaction space. We showed that the Ricci tensor components take the form

$$\begin{aligned} R_{00} &= -\text{Tr}(\dot{K}) - \text{Tr}(K^2), \\ R_{ij} &= E_{ij} + K_{ij}\text{Tr}(K). \end{aligned}$$

These expressions reveal that curvature is governed by two essential geometric ingredients:

1. The acceleration of the interaction metric (through $\dot{K}$),
2. The nonlinear self-coupling of the interaction flow (through K^2).

By defining the emotion quantity tensor

$$E_{ij} = \dot{K}_{ij} + K_{ik}K_j^k,$$

we obtain a geometrically consistent quantity describing intrinsic interaction acceleration. The temporal curvature component R_{00} corresponds to the negative trace of this interaction acceleration together with its nonlinear contribution, while the spatial components R_{ij} additionally include collective coupling through $\text{Tr}(K)$.

Therefore, curvature in this model is not imposed phenomenologically but emerges from the time evolution of the interaction metric itself. The geometry encodes how interaction structures deform, amplify, or stabilize over time.

This formulation establishes a mathematically consistent bridge between dynamical interaction systems and Riemannian curvature. It suggests that interaction dynamics may be interpreted as a geometric flow on a time-dependent manifold, opening possible connections to matrix Ricci flow, cosmological time-dependent metrics, and dynamical systems theory.

Further investigation may explore whether an interaction field equation—analogue to Einstein-type or Ricci flow equations—can be consistently formulated to govern the evolution of $G_{ij}(t)$.